ΠΑΝΕΠΙΣΤΗΜΙΟ ΙΩΑΝΝΙΝΩΝ
ΣΧΟΛΗ ΕΠΙΣΤΗΜΩΝ ΥΓΕΙΑΣ
Τμήμα Βιολογικών Εφαρμογών & Τεχνολογιών

Γενικά Μαθηματικά:* Τυπολόγιο

•Τριγωνομετρικές και υπερβολικές συναρτήσεις

$f(x) = \text{ημ}\, x = \sin x,\ x \in \mathbb{R},$

$f^{-1}(x) = \sin^{-1} x,\ x \in [-1,1],\ f^{-1}(x) \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right],$

$f(x) = \text{συν}\, x = \cos x,\ x \in \mathbb{R},$

$f^{-1}(x) = \cos^{-1} x,\ x \in [-1,1],\ f^{-1}(x) \in [0,\pi],$

$f(x) = \text{εφ}\, x = \tan x,\ x \in \left(k\pi - \frac{\pi}{2}, k\pi + \frac{\pi}{2}\right),\ k \in \mathbb{Z},$

$f^{-1}(x) = \tan^{-1} x,\ x \in \mathbb{R},\ f^{-1}(x) \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right],$

$f(x) = \text{σφ}\, x = \cot x,\ x \in\in (k\pi, k\pi + \pi),\ k \in \mathbb{Z},$

$f^{-1}(x) = \cot^{-1} x,\ x \in \mathbb{R},\ f^{-1}(x) \in [0,\pi],$

$\sec x = \frac{1}{\cos x},\ \operatorname{cosec} x = \frac{1}{\sin x}.$

$f(x) = \sinh x = \frac{e^x - e^{-x}}{2},\ x \in \mathbb{R},$

$f^{-1}(x) = \sinh^{-1} x = \ln\left(x + \sqrt{1+x^2}\right),\ x \in \mathbb{R},$

$f(x) = \cosh x = \frac{e^x + e^{-x}}{2},\ x \in \mathbb{R},$

$f^{-1}(x) = \cosh^{-1} x = \ln\left(x + \sqrt{x^2-1}\right),\ x \geq 1,$

$f(x) = \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}},\ x \in \mathbb{R},$

$f^{-1}(x) = \tanh^{-1} x = \frac{1}{2} \ln \frac{1+x}{1-x},\ |x| < 1,$

$f(x) = \coth x = \frac{1}{\tanh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}},\ x \in \mathbb{R} - \{0\},$

$f^{-1}(x) = \coth^{-1} x = \frac{1}{2} \ln \frac{1+x}{x-1},\ |x| > 1,$

$\operatorname{sech} x = \frac{1}{\cosh x},\ \operatorname{cosech} x = \frac{1}{\sinh x}.$

•Βασικές τριγωνομετρικές και υπερβολικές ταυτότητες

$\sin^2 x + \cos^2 x = 1,$

$\tan^2 x + 1 = \sec^2 x,$

$\cot^2 x + 1 = \operatorname{cosec}^2 x,$

$2\sec x = \sec x + \tan x + \frac{1}{\sec x + \tan x},$

$2\operatorname{cosec} x = \operatorname{cosec} x + \cot x + \frac{1}{\operatorname{cosec} x + \cot x},$

$\cosh^2 x - \sinh^2 x = 1,$

$1 - \tanh^2 x = \operatorname{sech}^2 x,$

$1 - \coth^2 x = -\operatorname{cosech}^2 x,$

$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y,$

$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y,$

$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y},$

$2\sin x \cos y = \sin(x+y) + \sin(x-y),$

$2\cos x \cos y = \cos(x+y) + \cos(x-y),$

$2\sin x \sin y = -\cos(x+y) + \cos(x-y),$

$\sin 2x = 2\sin x \cos x,$

$$\cos 2x = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x = 2\cos^2 x - 1,$$

$\tan 2x = \frac{2\tan x}{1 - \tan^2 x},$

$$\left.\begin{aligned} \sin x &= \frac{2t}{1+t^2},\ \cos x = \frac{1-t^2}{1+t^2}, \\ \tan x &= \frac{2t}{1-t^2}, \end{aligned}\right\}\ t = \tan\frac{x}{2}.$$

$\sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y,$

$\cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y,$

$\tanh(x \pm y) = \frac{\tanh x \pm \tanh y}{1 \pm \tanh x \tanh y},$

$2\sinh x \cosh y = \sinh(x+y) + \sinh(x-y),$

$2\cosh x \cosh y = \cosh(x+y) + \cosh(x-y),$

$2\sinh x \sinh y = \cosh(x+y) - \cosh(x-y),$

$\sinh 2x = 2\sinh x \cosh x,$

$$\cosh 2x = \cosh^2 x + \sinh^2 x = 1 + 2\sinh^2 x = 2\cosh^2 x - 1,$$

$\tanh 2x = \frac{2\tanh x}{1 + \tanh^2 x},$

$$\left.\begin{aligned} \sinh x &= \frac{2t}{1-t^2},\ \cosh x = \frac{1+t^2}{1-t^2}, \\ \tanh x &= \frac{2t}{1+t^2}, \end{aligned}\right\}\ t = \tanh\frac{x}{2}.$$

•Παράγωγοι και ολοκληρώματα βασικών συναρτήσεων

$\frac{\mathrm{d}}{\mathrm{d}x} a = 0,\ a:$ σταθερά,

$\frac{\mathrm{d}}{\mathrm{d}x} x^a = a x^{a-1},\ a \in \mathbb{R},$

περιορισμοί στο x ανάλογα με το a,

$\frac{\mathrm{d}}{\mathrm{d}x} a^x = a^x \ln a,\ a \in \mathbb{R}_+,\ x \in \mathbb{R},\ \frac{\mathrm{d}}{\mathrm{d}x} \mathrm{e}^x = \mathrm{e}^x,$

$\frac{\mathrm{d}}{\mathrm{d}x} \ln x = \frac{1}{x},\ x \in \mathbb{R}_+,$

$\frac{\mathrm{d}}{\mathrm{d}x} \sin x = \cos x,\ x \in \mathbb{R},$

$\frac{\mathrm{d}}{\mathrm{d}x} \cos x = -\sin x,\ x \in \mathbb{R},$

$\int 0\,\mathrm{d}x = a,\ \int a\,\mathrm{d}x = ax + c,\ a, c:$ σταθερές,

$\int x^a\,\mathrm{d}x = \frac{x^{a+1}}{a+1} + c,\ a \neq -1,$

$\int a^x\,\mathrm{d}x = \frac{a^x}{\ln a} + c,\ a \in \mathbb{R}_+ - \{1\},\ \int \mathrm{e}^x\,\mathrm{d}x = \mathrm{e}^x + c,$

$\int \frac{1}{x}\,\mathrm{d}x = \ln|x| + c,\ x \in \mathbb{R} - \{0\},$

$\int \sin x\,\mathrm{d}x = -\cos x + c,$

$\int \cos x\,\mathrm{d}x = \sin x + c,$

* *Διδάσκων*: Καθηγητής Κωνσταντίνος Αδαμίδης.

$$\frac{\mathrm{d}}{\mathrm{d}x}\tan x=\sec^2 x,\ x\in\left(k\pi-\tfrac{\pi}{2},k\pi+\tfrac{\pi}{2}\right),\ k\in\mathbb{Z},\qquad \int\tan x\,\mathrm{d}x=\ln|\sec x|+c,\quad \int\sec^2 x\,\mathrm{d}x=\tan x+c,$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\cot x=-\operatorname{cosec}^2 x,\ x\in(k\pi,k\pi+\pi),\ k\in\mathbb{Z},\qquad \int\cot x\,\mathrm{d}x=\ln|\sin x|+c,\quad \int\operatorname{cosec}^2 x\,\mathrm{d}x=-\cot x+c,$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\sec x=\sec x\tan x,\ x\in\left(k\pi-\tfrac{\pi}{2},k\pi+\tfrac{\pi}{2}\right),\ k\in\mathbb{Z},\qquad \int\sec x\tan x\,\mathrm{d}x=\sec x+c,\quad \int\sec x\,\mathrm{d}x=\ln|\sec x+\tan x|+c,$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\operatorname{cosec} x=-\operatorname{cosec} x\cot x,\ x\in(k\pi,k\pi+\pi),\ k\in\mathbb{Z},\qquad \int\operatorname{cosec} x\cot x\,\mathrm{d}x=-\operatorname{cosec} x+c,\quad \int\operatorname{cosec} x\,\mathrm{d}x=-\ln|\operatorname{cosec} x+\cot x|+c,$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin^{-1} x=\frac{1}{\sqrt{1-x^2}},\ |x|<1,\qquad \int\frac{\mathrm{d}x}{\sqrt{1-x^2}}=\sin^{-1}x+c,$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\cos^{-1} x=-\frac{1}{\sqrt{1-x^2}},\ |x|<1,$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\tan^{-1} x=\frac{1}{1+x^2},\ x\in\mathbb{R},\qquad \int\frac{\mathrm{d}x}{1+x^2}=\tan^{-1}x+c,$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\cot^{-1} x=-\frac{1}{1+x^2},\ x\in\mathbb{R},$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\sinh x=\cosh x,\ x\in\mathbb{R},\qquad \int\sinh x\,\mathrm{d}x=\cosh x+c,$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\cosh x=\sinh x,\ x\in\mathbb{R},\qquad \int\cosh x\,\mathrm{d}x=\sinh x+c,$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\tanh x=\operatorname{sech}^2 x,\ x\in\mathbb{R},\qquad \int\operatorname{sech}^2 x\,\mathrm{d}x=\tanh x+c,\quad \int\tanh x\,\mathrm{d}x=\ln(\cosh x)+c,$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\coth x=-\operatorname{cosech}^2 x,\ x\in\mathbb{R}-\{0\},\qquad \int\operatorname{cosech}^2 x\,\mathrm{d}x=-\coth x+c,\quad \int\coth x\,\mathrm{d}x=\ln|\sinh x|+c,$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\operatorname{sech} x=-\operatorname{sech} x\tanh x,\ x\in\mathbb{R},\qquad \int\operatorname{sech} x\tanh x\,\mathrm{d}x=-\operatorname{sech} x+c,$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\operatorname{cosech} x=-\operatorname{cosech} x\coth x,\ x\in\mathbb{R}-\{0\},\qquad \int\operatorname{cosech} x\coth x\,\mathrm{d}x=-\operatorname{cosec} x+c,$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\sinh^{-1} x=\frac{1}{\sqrt{1+x^2}},\ x\in\mathbb{R},\qquad \int\frac{1}{\sqrt{1+x^2}}\,\mathrm{d}x=\sinh^{-1}x+c,$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\cosh^{-1} x=\frac{1}{\sqrt{x^2-1}},\ x>1,\qquad \int\frac{1}{\sqrt{x^2-1}}\,\mathrm{d}x=\cosh^{-1}x+c,$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\tanh^{-1} x=\frac{1}{1-x^2},\ |x|<1,\qquad \int\frac{1}{1-x^2}\,\mathrm{d}x=\tanh^{-1}x+c,$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\coth^{-1} x=\frac{1}{1-x^2},\ |x|>1.\qquad \int\frac{1}{1-x^2}\,\mathrm{d}x=\coth^{-1}x+c,\quad \int\frac{1}{x^2-1}\,\mathrm{d}x=-\coth^{-1}x+c.$$

•Συναρτήσεις και ολοκληρώματα Γάμα και Βήτα, το τριώνυμο ως άθροισμα ή διαφορά τετραγώνων και το διωνυμικό ανάπτυγμα

$$\Gamma(x)=\int_0^\infty t^{x-1}\,\mathrm{e}^{-t}\,\mathrm{d}t,\ x>0,\qquad \mathrm{B}(m,n)=\int_0^1 x^{n-1}(1-x)^{m-1}=\frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)},$$

$$\Gamma(x+1)=x!=x(x-1)\times\cdots\times3\times2\times1,\ x\in\mathbb{N},\qquad \int_0^{\pi/2}\sin^{2m-1}\theta\cos^{2n-1}\theta\,\mathrm{d}\theta=\tfrac{1}{2}\,\mathrm{B}(m,n),$$

$$\Gamma(x+1)=x\Gamma(x),\ \Gamma(1)=1,\ x>0,\qquad \int_0^\infty\frac{x^{a-1}}{1+x}\,\mathrm{d}x=\mathrm{B}(a,1-a)=\frac{\pi}{\sin a\pi},\ a\in(0,1),$$

$$\Gamma(x)\Gamma(1-x)=\frac{\pi}{\sin x\pi},\ x\in(0,1),\ \Gamma\left(\tfrac{1}{2}\right)=\sqrt{\pi},\qquad \alpha x^2+\beta x+\gamma=-\frac{\Delta}{4\alpha}+\alpha\left(x+\frac{\beta}{2\alpha}\right)^2,\ \Delta=\beta^2-4\alpha\gamma,$$

$$\Gamma(2x)=\frac{2^{2x-1}\Gamma(x)\Gamma\left(x+\frac{1}{2}\right)}{\sqrt{\pi}},\qquad (a+b)^n=\sum_{k=0}^{n}\frac{n!}{k!(n-k)!}a^{n-k}b^k,\ n\in\mathbb{N},\ n\geq 2.$$